University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,010: IC: Strategic Management of New Technologies in Companies

ECTS credits: 4

Overview examination/s

(binding regulations see below)
decentral - Active participation, Analog, Individual work individual grade (25%)
Examination time: Term time
decentral - Presentation, Analog, Group work individual grade (50%)
Examination time: Term time
decentral - Presentation, Analog, Group work individual grade (25%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,010,1.00 IC: Strategic Management of New Technologies in Companies -- English -- Hein Andreas

Course information

Prerequisites

This course is assigned to the profile «Transforming and Managing Digital Business», but can also be taken without a selection of a specialization.

The course is open to students enrolled in the MBI and Master of Computer Science programs. There are no formal prerequirements.

Learning objectives

- Students engage with contemporary technological topics that are of strategic relevance to organizations. The primary goal is to identify real-world challenges at the intersection of technology and management and to investigate them through an academically rigorous research lens.
- Anchored in the tradition of **design science research** and **engaged scholarship**, the course provides a structured pathway for students to formulate and explore research questions that are both theoretically grounded and practically significant.
- Students work in teams—either independently or, optionally, in collaboration with industry partners—to address emerging topics such as the organizational deployment of artificial intelligence, platform transformation, or low-/no-code technologies.
- The overarching objective is to produce a **research-based project output**, which may serve as a foundation for a master's thesis or further academic work.
- The course supports students in developing their methodological and conceptual skills while providing opportunities to engage with real-world empirical material. This makes it particularly valuable for students considering an academic career or seeking to deepen their involvement in research-intensive roles in industry.

Content

The course focuses on the strategic application of emerging technologies in organizational contexts. Students are encouraged to develop research projects that address contemporary challenges and to investigate them through theory-informed empirical inquiry. Topics are chosen in consultation with the instructors and can be derived from personal interest, academic debate, or—in optional cooperation with partner firms—current industry problems.

The course is structured into **three on-site sessions** in St. Gallen, supplemented by self-directed work and individual coaching. Through this format, students are guided through a rigorous and iterative research process that emphasizes conceptual clarity, empirical grounding, and practical relevance.

1) Introduction and Problem Framing

The first session introduces students to the principles of design science research and agile research structuring. Conceptual foundations and methodological tools are presented to support topic selection, research design, and project planning. Contemporary research topics are outlined either by the teaching team or—optionally—by partner firms. Students form teams (typically 3–5 members), define their research focus, and begin scoping their project.

2) Interim Presentation and Feedback

In the second session, student teams present a 30-minute overview of their research progress, followed by a 15-minute discussion. Presentations should cover the research problem, theoretical framing, and initial empirical insights (if applicable). Constructive feedback from the teaching team—and where relevant, from external partners—supports iterative refinement.

3) Final Presentation

The course concludes with 30-minute final presentations and a 15-minute discussion. Each team presents their completed research project, demonstrating a coherent narrative from problem identification to research execution and findings. Presentations should reflect academic rigor, theoretical contribution, and practical relevance. A separate written report is not required; instead, the presentation serves as the central deliverable.

Structure and indications of the learning and teaching design

The course carries **4 ECTS credits**, equivalent to a **total workload of 120 hours**, including contact sessions, self-study, and assessments.

The course follows a **research-intensive and practice-relevant format**. Students work in teams to develop a research project on a timely technological topic. Collaboration with industry partners is optional. The final assessment is based on:

- **Interim Presentation** (25%, group grade)
- **Final Presentation** (50%, group grade)
- **Active Engagement and Participation** (25%, individual grade)

There is **no separate written report**. Instead, students are expected to develop a research presentation that integrates theory, empirical exploration, and critical reflection.

On-Campus Sessions

(All session dates will be announced on **StudyNet**)

- **Introductory Session**
  3-hour session introducing methodological foundations, topic selection, and team formation.
- **Interim Presentations**
  Each team presents preliminary findings (30 minutes + 15 minutes discussion). Feedback informs project refinement.
- **Final Presentations**
  Teams present their final results (30 minutes + 15 minutes discussion), demonstrating how their research addressed the identified challenge.

Self-Directed Learning

- **Preparation for Presentations (~70 hours):**
  Literature review, project coordination, empirical inquiry (if applicable), and development of presentations.

- **Personal Coaching (up to 5 hours):**
  Teams may schedule individual coaching sessions with the instructor to support research development and refinement.

Location

- All sessions will take place in person in **St. Gallen**. The course fosters a collaborative academic environment through direct interaction with instructors and peers. Final session dates and logistical details will be communicated via **StudyNet**.

Literature

Baskerville, R., Baiyere, A., Gregor, S., Hevner, A., & Rossi, M. (2018). Design science research contributions: Finding a balance between artifact and theory. Journal of the Association for Information Systems, 19(5), 3.

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. MIS Quarterly, 37(2), 337-355.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75-105.

Hevner, A. R. (2007). A three cycle view of design science research. Scandinavian journal of information systems, 19(2), 4.

Rogers, E. M. (1962). Diffusion of Innovations. Free Press.; Moore, G. A. (1991). Crossing the Chasm: Marketing and Selling High-Tech Products to Mainstream Customers. Harper Business.

Oswald G, Krcmar H (2018) Digitale Transformation – Fallbeispiele und Branchenanalysen, 1. Aufl. Springer Gabler, Garching.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. Journal of management information systems, 24(3), 45-77.

Tuunanen, T., Winter, R., & Vom Brocke, J. (2024). DEALING WITH COMPLEXITY IN DESIGN SCIENCE RESEARCH: A METHODOLOGY USING DESIGN ECHELONS. MIS Quarterly, 48(2).

Additional information

--

Examination information

Examination sub part/s

1. Examination sub part (1/3)

Examination modalities

Examination type	Active participation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus

Grading type	Individual work individual grade
Weighting	25%
Duration	--

Languages
Question language: English
Answer language: English

Remark
Active Participation

Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
--

2. Examination sub part (2/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work individual grade
Weighting	50%
Duration	--

Languages
Question language: English
Answer language: English

Remark
Final Presentation

Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
--

3. Examination sub part (3/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral

Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work individual grade
Weighting	25%
Duration	--

Languages
Question language: English
Answer language: English

Remark
Interim Presentation

Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
--

Content

The examination in this course is based on the development and presentation of an original, research-oriented project that addresses a contemporary technological challenge relevant to organizations. The assessment focuses on the student teams' ability to formulate a theoretically grounded research question, design a coherent inquiry process, and derive academically and practically meaningful insights.

The following subject areas and competencies are relevant to the examination:

1. Problem Framing and Topic Selection

- Ability to identify a strategically relevant and timely technological issue in an organizational context (e.g., AI deployment, platform transitions, automation).
- Capacity to articulate a clear research problem that aligns with both academic debates and real-world relevance.
- Familiarity with engaged scholarship and design science research as guiding approaches.

2. Theoretical and Conceptual Grounding

- Competence in selecting and applying appropriate theoretical frameworks (e.g., IS theory, strategic management, digital transformation).
- Understanding how to position the research within relevant academic discourse.
- Clarity and coherence in the formulation of research objectives and assumptions.

3. Research Design and Methodology

- Understanding of research structuring techniques, including iterative, agile approaches to project development.

- Ability to select and justify appropriate empirical methods (e.g., case study design, interviews, document analysis).
- Reflection on methodological rigor and validity.

4. Empirical Engagement (where applicable)

- Skill in gathering and analyzing empirical data relevant to the research question.
- Integration of empirical insights with theoretical arguments.
- Critical reflection on the scope and limitations of the data.

5. Synthesis and Presentation of Results

- Coherent development of arguments and derivation of findings that address the research question.
- Ability to link theoretical frameworks, empirical insights, and managerial implications.
- Effective communication of research outcomes in a professional presentation format.

6. Active Participation and Reflective Engagement

- Regular contribution to team discussions and coaching sessions.
- Demonstrated progress between interim and final presentations.
- Openness to feedback and ability to incorporate it into project refinement.

The **interim presentation** serves as a formative assessment focused on early-stage conceptual clarity, research design, and project feasibility. The **final presentation** constitutes the summative assessment and reflects the maturity, coherence, and contribution of the overall research effort. Active participation is assessed continuously throughout the course, with attention to individual engagement, preparation, and collaboration.

Literature

See literature above.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).